

Mini-Project 2

Homework No. 2

ChE 396 - Senior Design I

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Project Charter 1

Title	Goal	In Scope	Out of Scope
Mini Project 2	Optimization of utility consumption by heat integration (heat exchanger network)	• Create PFDs before and after heat integration • Create composite curves • Explain how to find ΔT_{min} optimized • Calculate pinch temperature • Create heat exchanger network design • Utility costs before and after heat integration	• Creating a heat exchanger design • Utility costs other than those used for heating and cooling
Deliverables		Team Members	
• PFD before and after heat integration • Composite curves • Calculating optimized ΔT_{min} explanation • Pinch Temperature calculation • Heat exchanger network design explanation • Utility costs before and after heat integration calculations		Mahamat Tahir Abdraman Diana Awimrin Jon Chiu Yaser Khalid	

Design Plan 1

*** We all helped each other with all the tasks. The names written next to each task are just the people in charge of most of that section to keep us on track while working. ***

Mini Project 2					
Task ID	Task	Start Date	Duration (Days)	Finish Date	Assignee
Logistics		3-Oct	1	4-Oct	
1	First meeting (Task Delegation)	3-Oct	1	4-Oct	All
2	Project Charter	3-Oct	1	4-Oct	Diana
3	Design Plan	3-Oct	1	4-Oct	Diana
4	Layout/Format	3-Oct	1	4-Oct	All
Execution and Process Description		4-Oct	4	7-Oct	
5	PFD Before and After Heat Integration	4-Oct	1	4-Oct	Jon
6	Composite Curves	4-Oct	1	4-Oct	Diana
7	Calculating Optimized ΔT_{min}	5-Oct	1	5-Oct	Diana
8	Pinch Temperature Calculation	5-Oct	1	5-Oct	Diana
9	Heat Exchanger Network Design	6-Oct	1	6-Oct	Yaser
10	Utility Costs Before and After Heat Integration	7-Oct	1	7-Oct	Yaser
Wrap Up		10-Oct	1	10-Oct	
11	Section Unification	10-Oct	1	10-Oct	All
12	Proofreading/Plagiarism Check	10-Oct	1	10-Oct	All

Problem 1

PFD's Before and After Heat Integration

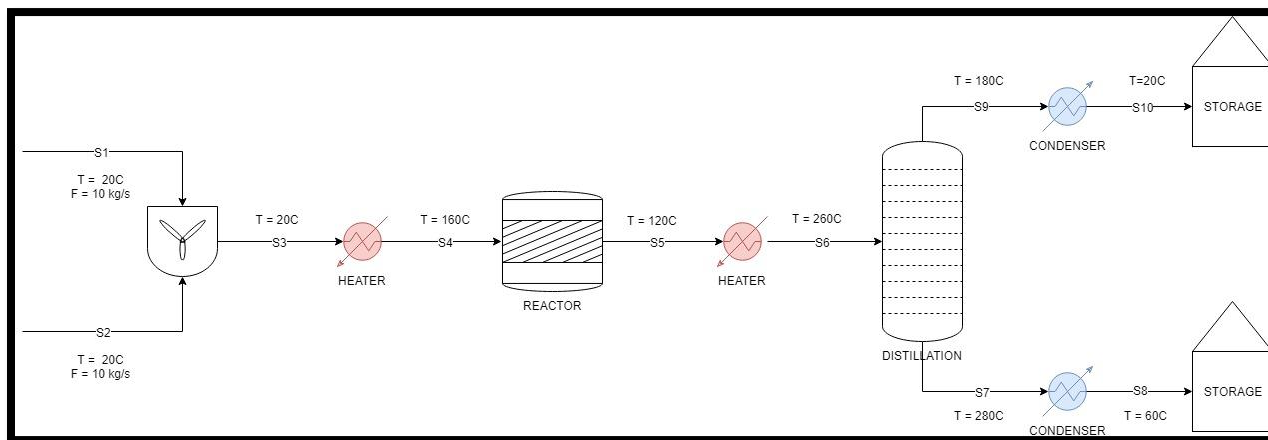


Figure 1: PFD Before Heat Integration

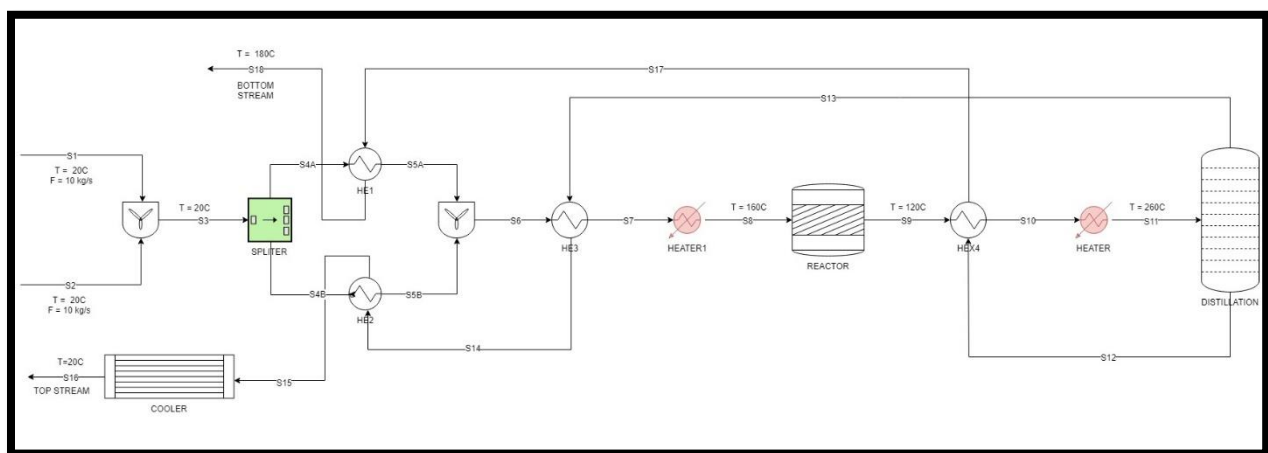


Figure 2: PFD After Heat Integration

Composite Curves

Composite curves are important to compile because they are used to find several important temperatures and enthalpies in the process. The pinch temperature, the optimized ΔT_{min} , and the enthalpies of both the hot and cold utilities can be determined from the composite curve. In order to plot the data for a combined composite curve of both the cold and hot stream, a couple preliminary composite curves need to be plotted using the information shown in Table 1 below.

Table 1: Stream Data

Stream #	Temperature (°C)		Heat Capacity Flow Rate CP (kW/°C)	Heat Load (kW)
	T _{in}	T _{out}		
Cold 1	20	160	50	7000
Cold 2	120	260	55	7700
Hot 1	180	20	40	6400
Hot 2	280	60	30	6600

First, a composite curve for the cold streams is created. This is done by plotting several temperatures going from 20 °C to 160 °C for the y-axis. Next, the enthalpies for the x-axis are determined by multiplying each temperature by the heat capacity flow rate for the stream which is 50 kW/°C. The same procedure is followed for the second cold stream. After that, each of the streams respective data is plotted onto one graph to result in Figure 3 shown below.

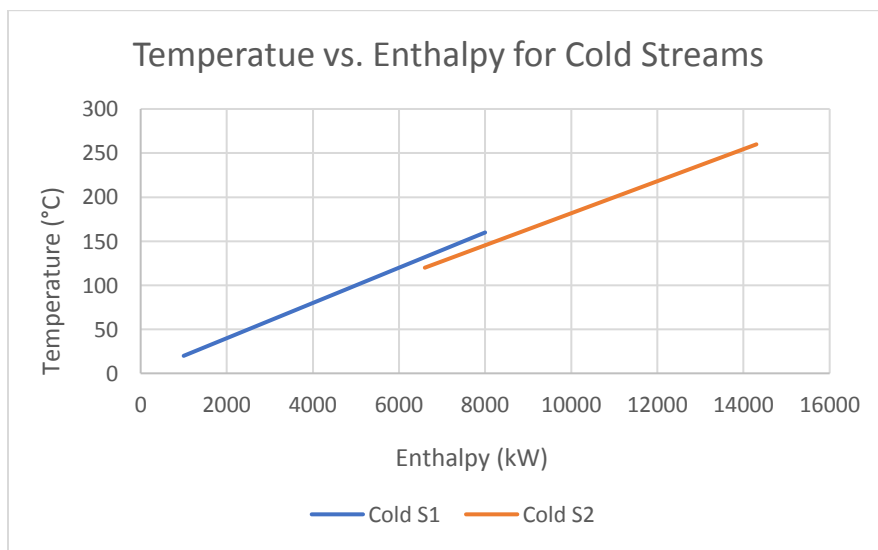


Figure 3: Cold Streams' Composite Curve (not combined)

Table 2 Cold Streams' Temperatures and Enthalpies

T (°C)	H (kW)	T (°C)	H (kW)
20	1000	120	6600
35	1750	135	7425
50	2500	150	8250
65	3250	160	8800
80	4000	175	9625
95	4750	190	10450
110	5500	205	11275
120	6000	220	12100
135	6750	235	12925
150	7500	250	13750
160	8000	260	14300

To combine the two streams together, a point slope calculation is carried out where there is an area between the lines overlapping. The first point is at 120 °C and 6000 kW and the second point is at 160 °C and 8800 kW and results in a combined cold streams' composite curve shown in Figure 4 below.

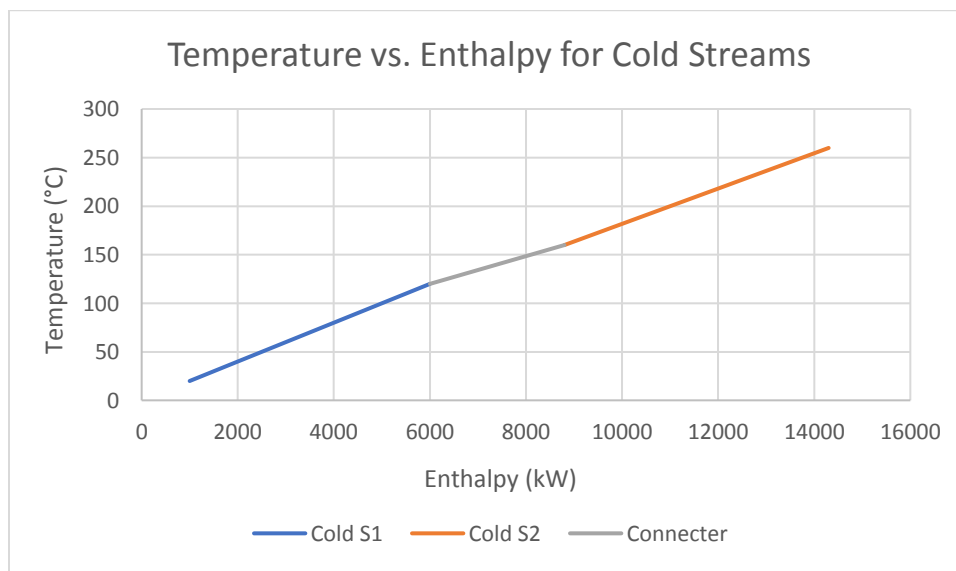


Figure 4: Cold Streams' Composite Curve (combined)

The same procedure explained above is repeated for the hot streams resulting in the graphs shown in Figure 5 and Figure 6 below.

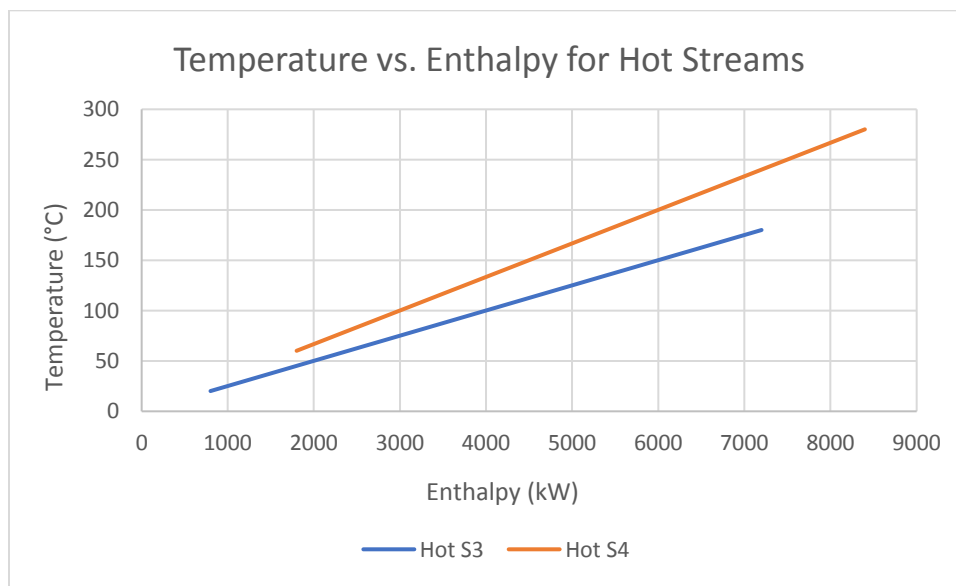


Figure 5: Hot Streams' Composite Curve (not combined)

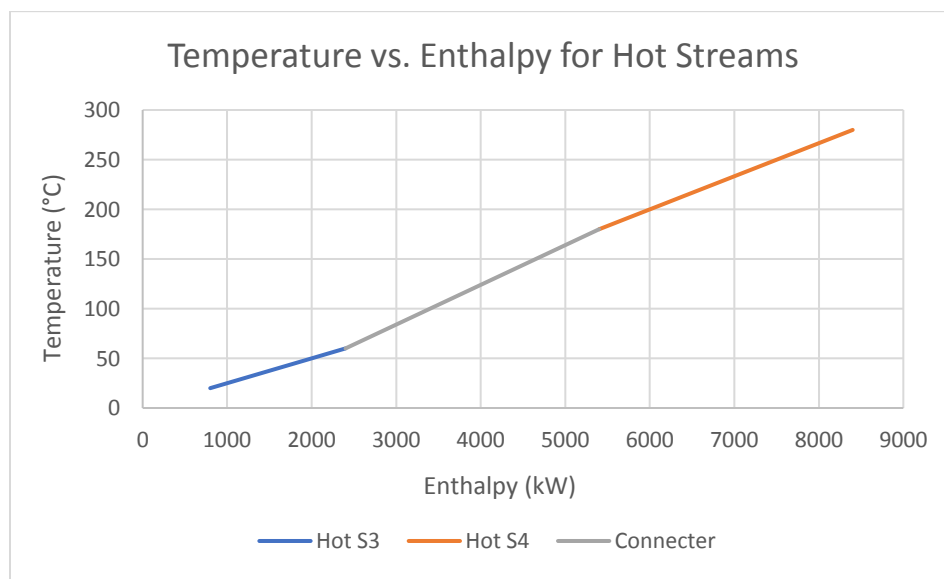


Figure 6: Hot Streams' Composite Curve (combined)

Next, the two “combined” graphs are plotted into one graph (shown below in Figure 7) and can be used to determine the aforementioned information such as the pinch temperature, the optimized ΔT_{min} , and the enthalpies of both the hot and cold utilities.

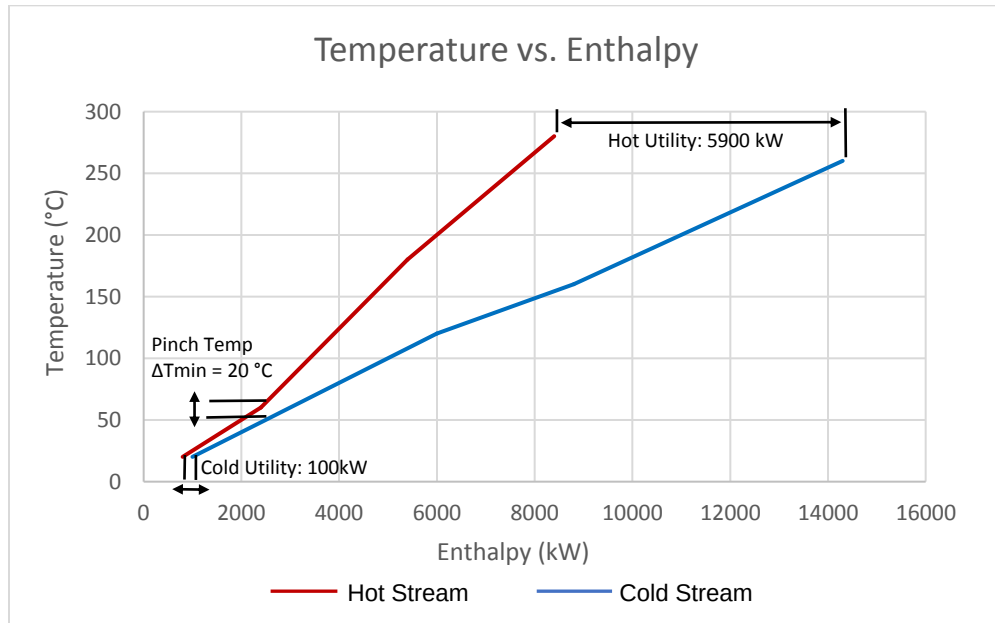
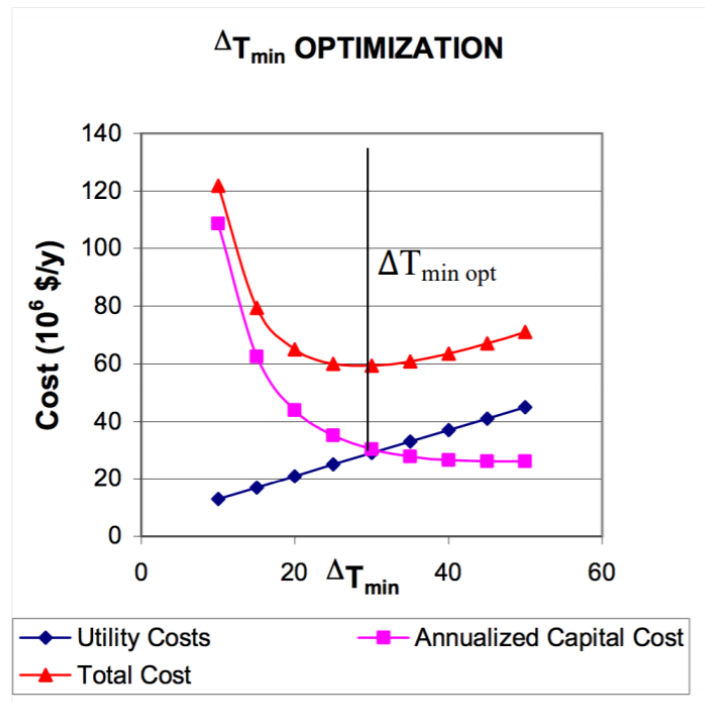


Figure 7: Hot and Cold Composite Curve (combined)

Calculating Optimized ΔT_{min}

Calculating the optimized ΔT_{min} of a process requires a pinch analysis and composite curves to be plotted before any further steps. To find the optimized ΔT_{min} , a Total Annualized Cost vs. ΔT_{min} plot needs to be created as shown in Table 3 below [2]. In order to do this, determining the capital costs and operating costs for a range of ΔT_{min} values is crucial to the procedure. These two costs are then added together to give an annualized cost. The data of all the utility costs, annualized capital cost, and total cost for the process at varying ΔT_{min} values are then plotted. With all these trend lines placed on one graph, the next step is to pinpoint the intersection between the utility cost and the annualized capital cost curves. That would be the where the ΔT_{min} is supposedly most optimized. After this is complete, going back to revise the process would be necessary. Many factors can affect ΔT_{min} such as the amount and size of heat exchangers used in the process and the addition or subtraction of heating/cooling streams. Any changes done to the process will then prompt in repeating the steps of making new composite curves and a Total Annualized Cost vs. ΔT_{min} plot until the most optimized ΔT_{min} is found. As seen, the way the process is carried out most definitely affects the optimization of ΔT_{min} and needs to be considered carefully.

Table 3: Optimization of ΔT_{min} [2]



Pinch Temperature Calculation ($\Delta T_{min} = 20^\circ\text{C}$)

Pinch Temperature can be solved in three different ways: through creating the composite curves, through hand calculations, or using a pinch analysis excel template found online [2]. To solve this problem, an excel template will be used because it is the most efficient way to obtain the temperature. To solve for the pinch temperature some vital data about the streams needs to be collected and plugged into the “Problem-Table” excel sheet [2]. First, the initial and final temperatures of the hot and cold streams are plugged into the excel sheet. Then, the C_p values given in units of $\text{kJ/kg } ^\circ\text{C}$ are converted into $\text{kW/}^\circ\text{C}$ by multiplying each one by its respective flow rate given in kg/s . The C_p values of the cold streams going from 20°C to 160°C and 120°C to 260°C are both multiplied by 10 kg/s . The distillate hot stream going from 180°C to 20°C , has a 40% mass so the C_p value given ($10 \text{ kJ/kg } ^\circ\text{C}$) is multiplied by 4 kg/s . The same is done for the bottoms hot stream going from 280°C to 60°C , but with a 60% mass. The rest of the excel sheet is then auto filled out and the pinch temperature is identified by finding the point of convergence for the heat load which is between 1400 kW and 1600 kW .

The pinch temperature is found to be 130 °C.

Table 4: Problem-Table Calculating Pinch Temperature [3]

1. Minimum temperature approach

$$\Delta T_{\min} = 20 \text{ }^{\circ}\text{C}$$

2. Stream data

Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)
	Source	Target	Source	Target		
1	20	160	30	170	50	7000
2	120	260	130	270	55	7700
3	180	20	170	10	40	6400
4	280	60	270	50	30	6600
5	0	0	10	10	0	0
6	0	0	10	10	0	0
7	0	0	10	10	0	0
8	0	0	10	10	0	0

3. Problem table

Interval	Interval temp (°C)	Interval ΔT (°C)	Sum CP c - sum CP h (kW/°C)	dH (kW)	Cascade (kW)	(kW)
1	270	0	0	0	0	3900
2	270	0	0	0	0	3900
3	170	100	25	2500	-2500	1400
4	170	0	25	0	-2500	1400
5	130	40	35	1400	-3900	0
6	50	80	-20	-1600	-2300	1600
7	30	20	10	200	-2500	1400
8	10	20	-40	-800	-1700	2200

Heat Exchanger Network Design

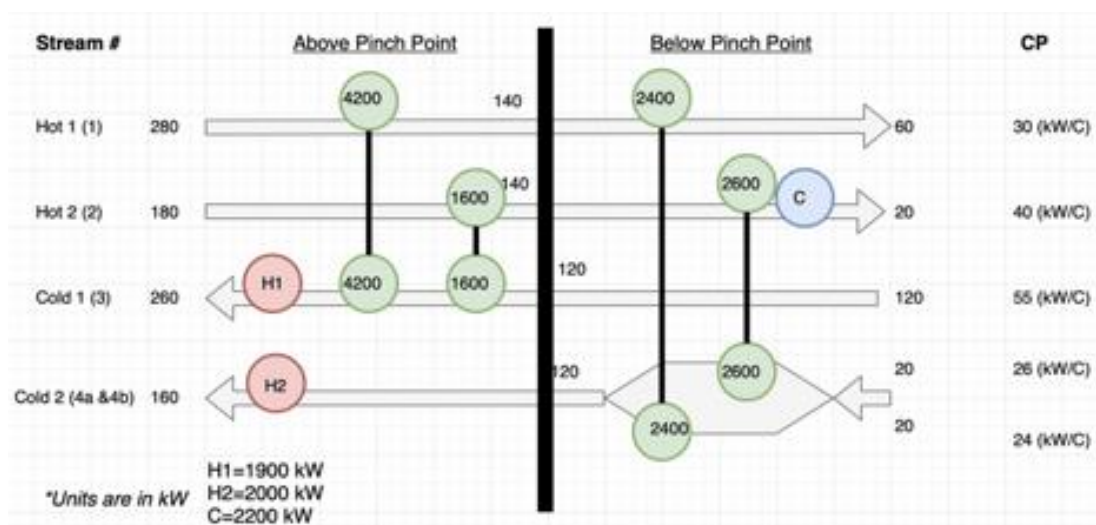


Figure 8: Heat Exchanger Network Design

Above Pinch Point

$$CP_{\text{Hot}} \leq CP_{\text{Cold}}$$

Stream 2 to Stream 3

$$\text{Stream 2: } 40 * (180 - 180) = 1600 \text{ kW}$$

$$\text{Stream 3: } 55 * (260 - 120) = 7700 \text{ kW}$$

Stream 1 to Stream 3

$$\text{Stream 1: } 30 * (280 - 140) = 4200 \text{ kW}$$

Stream 3 required hot utility of 1900 kW

$$7700 \text{ kW} - (1600 \text{ kW} + 4200 \text{ kW}) = 1900 \text{ kW}$$

Stream 4 required hot utility of 2000 kW

$$\text{Stream 4: } 50 * (160 - 120) = 2000 \text{ kW}$$

Below Pinch Point

$$CP_{\text{Hot}} \geq CP_{\text{Cold}}$$

Stream 1 to Stream 4b (Countercurrent)

$$\text{Stream 1: } 30 * (140 - 60) = 2400 \text{ kW}$$

$$\text{Stream 4b: } 24 * (120 - 20) = 2400 \text{ kW}$$

Stream 4A to Stream 2

$$\text{Stream 4A: } 26 * (120 - 20) = 2600 \text{ kW}$$

$$\text{Stream 2: } 40 * (140 - 20) = 4800 \text{ kW}$$

Stream 2 required cold utility of 2200 kW

$$4800 - 2600 = 2200 \text{ kW}$$

Above the pinch point, the CP for hot streams must be less than the CP for cold stream to be able to match. Stream 1 and stream 2 have a lower CP than stream 3, therefore they can match. The load from hot and cold do not counterbalance each other, requiring a heat utility to be placed on stream 3. Stream 4 will also need its own heat utility as it could not match with any hot streams. Below the pinch, the CP of cold streams now must be lower than the CP of hot streams to match. Stream 4 is split into two, with the stream 4a matching stream 2, and stream 4b matching with stream 1. Stream 1 and stream 4b will counterbalance each other's load, therefore requiring no cold utility. Stream 2 will need a cold utility as its load is higher than stream 4a's.

One thing to note is that stream 3 is at T_{Cold} temperature, therefore does not require any heat transfers. Hot utilities are only put above the pinch point and cold utilities are only put below the pinch point.

Utility Costs Before and After Heat Integration

Table 5: Cost of Utilities Before Heat Integration

Stream	Temp	Utility	Flowrate	CP(kJ/kg*°C)	Cp(Kw*°C)	ΔH (kW)
s1	20		10	5		
s2	20		10	5		
s3	20	Utility 1	20	5	100	14000
s4	160		20	5	100	
s5	120	Utility 2	20	5.5	110	15400
s6	260		20	5.5	110	
s7	180			10		
s8	180	Utility 3	8	10	80	12800
s10	20		8	10	80	
s9	180			10		
s11	280			5		
s12				5		
s13	280	Utility 4	12	5	60	13200
s14	60		12	5	60	

Table 6: Cost of Utilities Before Heat Integration

Before Heat Integration	
Utility	Operating Costs (\$/yr)
1	\$10,792,320.00
2	\$11,871,552.00
3	\$9,867,264.00
4	\$10,175,616.00
Total	\$42,706,752.00
Price of Power (\$/kW)	\$0.11
Utility Efficiency	0.8
Operation hr/yr	8760

Table 7: Cost of Utilities After Heat Integration

After Heat Integration	
Utility	Operating Costs (\$/yr)
Cold	\$1,695,936.00
Hot	\$3,006,432.00
Total	\$4,702,368.00

Both tables above show the cost of utilities for before and after heat integration, respectively. The utilities chosen are steam for heat utility and cooling water for cold utility. Price per kW chosen is the average price per kilowatt in Illinois. Utility efficiently chosen is an optimal 80 percent. The equation used to calculate utility costs is as follows:

$$Utility\ Costs = \Delta H(kW) * Price\ of\ Power\left(\frac{\$}{kW}\right) * Utility\ Efficiency * Operating\ Period$$

Project Charter 2

Title	Goal	In Scope	Out of Scope
Mini Project 2	Explain the process of producing MTBE and simulate one of the distillation columns	• Creating BFD of process • Feeds and mass balances for each stage • PFD for process • Simulation of MTBE Column with Aspen	• Heat Exchanger Design • Aspen simulation for the entire process
Deliverables		Team Members	
• Block flow diagram for this process • Feeds and mass balance for each stage • PFD for the process • Simulation of MTBE Column with Aspen		Mahamat Tahir Abdraman Diana Awimrin Jon Chiu Yaser Khalid	

Design Plan 2

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Execution and Process Description		4-Oct	6	9-Oct	
5	Block Flow Diagram	4-Oct	1	4-Oct	Jon
6	Feeds and Mass Balances for Each Stage	5-Oct	2	6-Oct	Jon
7	PFD	7-Oct	1	7-Oct	Jon
8	Simulation of MTBE Column with Aspen	8-Oct	2	9-Oct	Mahamat
Wrap Up		10-Oct	1	10-Oct	
9	Section Unification	10-Oct	1	10-Oct	All
10	Proofreading/Plagiarism Check	10-Oct	1	10-Oct	All

Problem 2

Process Description

The process of producing Methyl tertiary butyl ether (MTBE) involves the reaction of isobutene with methanol. Methanol and mixed butylene are first mixed and preheated before being pumped into reactor at desired operating conditions. The product containing MTBE, methanol, C4 compounds and unimportant side reactions are sent to a drum where methanol and C4 compounds leave at the top of the column as vapor and MTBE at the bottom as pure liquid. The methanol and C4 compounds are then sent to an absorption tower where methanol is separated using water solvent and C4 compounds leave at the top as fuel gas. The unreacted methanol is further separated from water using a distillation tower where methanol leaves at the top and is recycled back to reactor feed and water solvent leaves at the bottom and recycled back to the absorption tower. A purge stream is taken from the water before it is recycled to minimize impurities build up. See figure below for process.

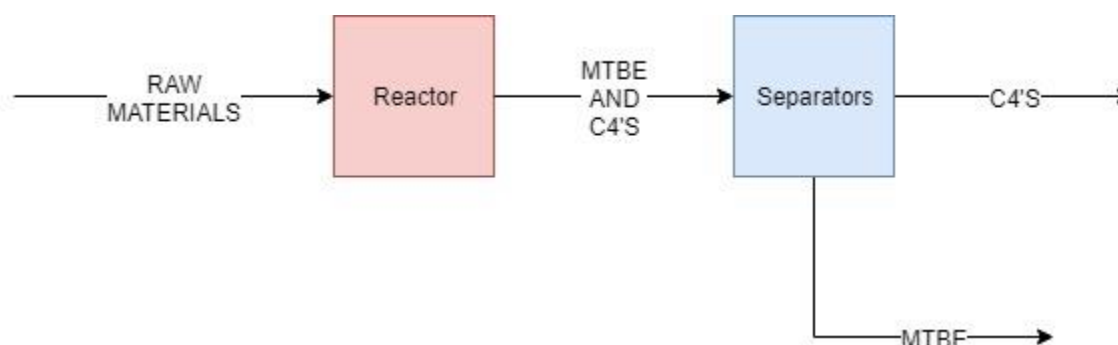


Figure 9 BFD of MTBE production

Distillation is a process of purifying a liquid by heating and cooling, in term, it is a separation of two or more components by the difference of their boiling point temperature, as result, the component with a lower boiling point will evaporate first and become the distillate product. However, a strange occurrence may observe for the separation process between methanol and MTBE in a distillation. According to PubChem, the boiling points for methanol is 64.7 °C while the MTBE is 55.2 °C [4], but the former becomes the distillate product while the latter becomes the bottom product in a distillation process. This occurrence was due to the boiling points and the volatilities of the components being altered by the pressure, and resulted methanol lower its boiling points when pressure raised above 3 bars [4], and as PubChem stated that the boiling point for methanol was taken in atmospheric pressure, essentially the concept of boiling points manipulation was founded on the ideal gas law, $PV = nRT$, with a the variations that based on volatilities and compressibility. [1][4]

Material Balance for MTBE Production

The basis for this design is to produce 7000 kg/hr of MTBE via distillation in a steady state process, and the recovery rate is 99.5% in S5. As the lynch pin to this whole process, convert the MTBE to kmol/hr while reverse the 99.5% to calculate the MTBE content in the inlet stream S4.

$$7000 \frac{kg}{hr} * \frac{1}{88.15} \frac{kmol}{kg} * \frac{1}{99.5\%} = 79.81 \frac{kmol}{hr} MTBE$$

The Stoichiometry for the reaction is 1:1 ratio for the methanol to iso-butylene with a conversion rate of 97%. Reverse the reaction conversion to determine the amount of iso-butylene required before the reactor ins S3.

$$79.81 \frac{kmol}{hr} * \frac{1}{97\%} = 82.28 \frac{kmol}{hr} isobutylene$$

The amount of methanol in S3 required for the conversion is 110% of the iso-butylene to suppresses the side reaction.

$$82.28 \frac{kmol}{hr} * 110\% = 90.51 \frac{kmol}{hr} methanol$$

The content of iso-butylene and the C4s in S1 are identical to S3 while the C4s can be calculated based on the percentage of Isobutylene with their respective component mol%. The C4s can further combined as a single component for easier future calculation, as described in the table 8 below.

Spices	Mass Flow Rate	Units	Molar Flow Rate	Units	Mol%
Isobutylene	4616.589	kg/hr	82.27747	kmol/hr	49%
1-Butene	2920.699	kg/hr	52.05309	kmol/hr	31%
2-Butene	1695.89	kg/hr	30.22438	kmol/hr	18%
n-Butane	195.1823	kg/hr	3.358264	kmol/hr	2%
C4s sum	4811.771	kg/hr	85.63573	kmol/hr	

Table 8 Calculated components in S2

With the given information, the point of interest shifted back to Distillation Column 1. Since the recovery rate for MTBE was 99.5%, then the amount of methanol on S5 equals 0.5% of the MTBE on S4. Additionally, since the C4s components and the Isobutylene have a very low boiling points that ranged between -6.9°C to 3.72° [4], assumption was made that the C4s and Isobutylene were 0% in S5. Subsequently, subtracting S5 from S4, the flowrate of individual components in S6 can be calculated, as presented in table 10 and 11.

The next point of interest is the Absorption Column. 99% of the methanol from S6 were recovered, with the molar concentration of 15% when absorbed by water, then the mixture was exiting the column through S7. With the given information, the water molar flow rate required

for the absorption of methanol can be calculated. Subsequently, subtracting the components of S7 from the components of S6, the flowrate of the components in S8 can be calculated.

$$10.19 \frac{\text{kmol}}{\text{hr}} * \frac{85\%}{15\%} = 57.77 \frac{\text{kmol}}{\text{hr}} \text{ Water}$$

As S7 enters Distillation Column 2, another assumption was made based on the given specification, the pressure in distillation column 2 essentially operates under atmospheric pressure, and water's boiling point temperature is more than 20°C higher than methanol [4], thus 100% separation of methanol and water can be achieved, and they should exit through their respective streams where methanol recycles back to the mixer through S10, and water leaves the distillation column through S11.

A Splitter is installed to purge out 10% of the water from S11 through S13 in case of impurity build up, and 90% are recycled back to the absorption column through S12. To keep a steady state for the water in the absorption column, inlet must equal to the outlet, thus the inlet flowrate of S9 will equal to the outlet flowrate S13.

To complete the cycle for this distillation process, subtract the methanol component of the recycle stream S10 from the methanol component of S3 to determine the initial feed of methanol required from S2. The material balances are calculated in molar flowrate then converted to mass flowrate, as presented in follow figures and data tables.

Mixer:

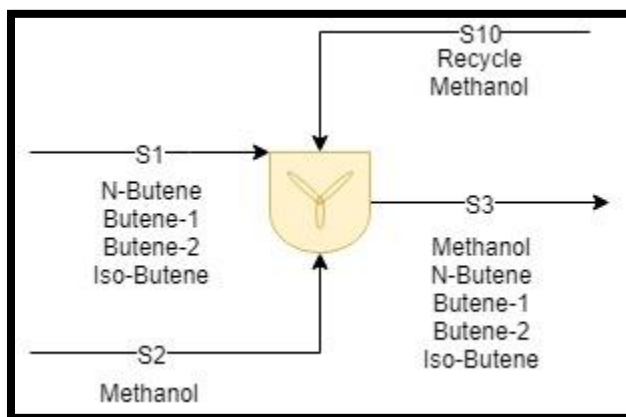


Figure 10 Inlet and Outlet stream of the Mixer

Table 9 Calculated material balance for the Mixer.

S1	Mass Flow Rate	Units	Molar Flow Rate	Units
Isobutene	4616.589	kg/hr	82.27747	kmol/hr
C4's	4811.771	kg/hr	85.63573	kmol/hr
S1 total	9428.36	kg/hr	167.9132	kmol/hr
S2	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	2573.17	kg/hr	80.31116	kmol/hr
S2 total	2573.17	kg/hr	80.31116	kmol/hr
S10	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	326.6175	kg/hr	10.19405	kmol/hr
S10 total	326.6175	kg/hr	10.19405	kmol/hr

Reactor:

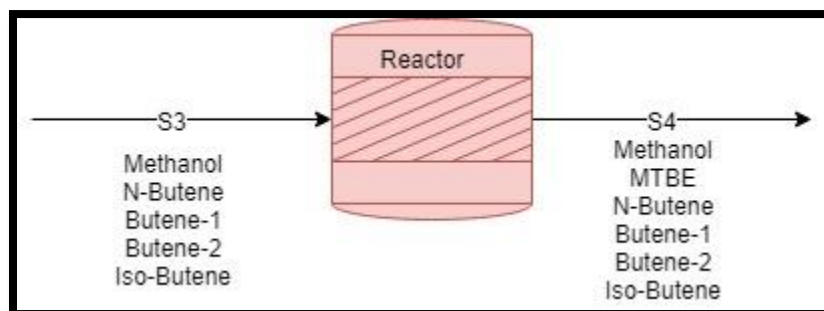


Figure 11 Inlet and Outlet Stream of Reactor

Table 10 Calculated material balance for the Reactor

S3	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	2899.787	kg/hr	90.50521	kmol/hr
Isobutene	4616.589	kg/hr	82.27747	kmol/hr
C4's	4811.771	kg/hr	85.63573	kmol/hr
S3 total	12328.15	kg/hr	258.4184	kmol/hr
S4	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	342.7021	kg/hr	10.69607	kmol/hr
MTBE	7035.176	kg/hr	79.80914	kmol/hr
Isobutene	138.4977	kg/hr	2.468324	kmol/hr
C4's	4811.771	kg/hr	85.63573	kmol/hr
S4 total	12328.15	kg/hr	178.6093	kmol/hr

Distillation Column 1:

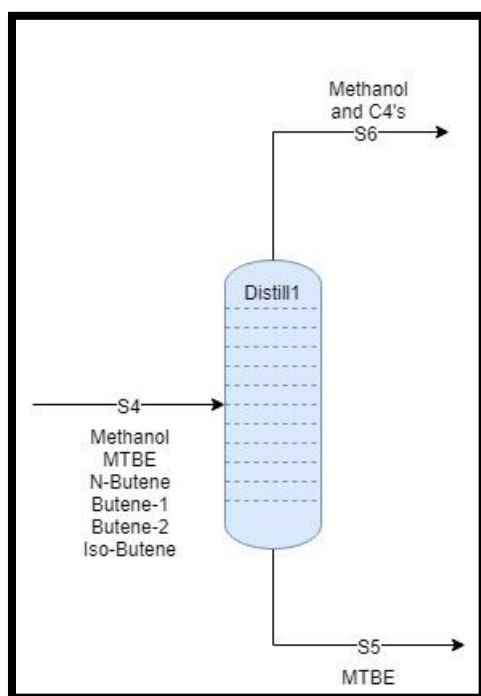


Figure 12 Inlet and Outlet Stream of Distillation Column 1

Table 11 Calculated material balance for Distillation Column 1

S5	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	12.78542	kg/hr	0.399046	kmol/hr
MTBE	7000	kg/hr	79.4101	kmol/hr
Isobutene	-	kg/hr	-	kmol/hr
C4's	-	kg/hr	-	kmol/hr
S5 total	7012.785	kg/hr	79.80914	kmol/hr
S6	Mass Flow Rate	Units	Molar Flow Rate	Units
Methanol	329.9167	kg/hr	10.29702	kmol/hr
MTBE	35.17588	kg/hr	0.399046	kmol/hr
Isobutene	138.4977	kg/hr	2.468324	kmol/hr
C4's	4811.771	kg/hr	85.63573	kmol/hr
S6 total	5315.361	kg/hr	98.80012	kmol/hr

Absorption Column:

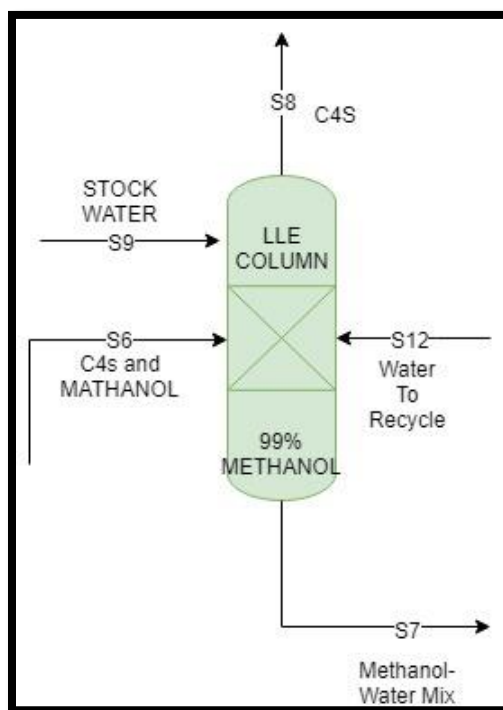


Figure 13 Inlet and Outlet Stream of Absorption Column

Table 12 Calculated material balance for Absorption Column

S7	Mass Flow Rate		Molar Flow Rate	Units
Methanol	326.6175	kg/hr	10.19405	kmol/hr
Water	1040.66	Kg/hr	57.76631	kmol/hr
S7 total	1367.278	kg/hr	67.96036	kmol/hr
S8	Mass Flow Rate		Molar Flow Rate	Units
Methanol	3.299167	kg/hr	0.10297	kmol/hr
MTBE	35.17588	kg/hr	0.399046	kmol/hr
Isobutene	138.4977	kg/hr	2.468324	kmol/hr
C4's	4811.771	kg/hr	85.63573	kmol/hr
S8 total	4988.744	kg/hr	88.60607	kmol/hr
S9	Mass Flow Rate		Molar Flow Rate	Units
Water	104.066	kg/hr	5.776631	kmol/hr
S9 total	104.066	kg/hr	5.776631	kmol/hr

Distillation Column 2 and Splitter:

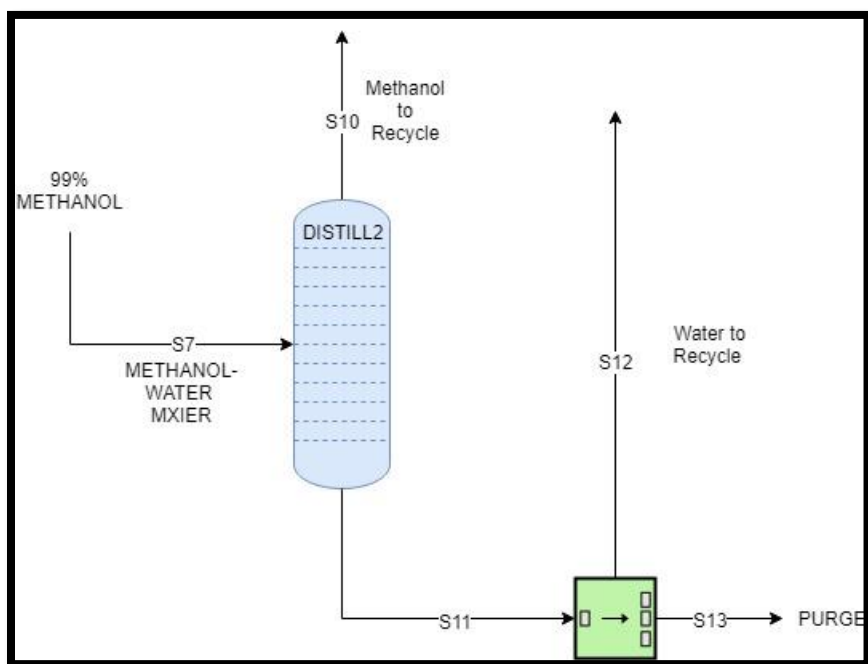


Figure 14 Inlet and Outlet Stream of Distillation Column 2 with Splitter.

Table 13 Calculated material balance for Distillation Column 2 and Splitter

S11	Mass Flow Rate	Units	Molar Flow Rate	Units
Water	1040.66	kg/hr	57.76631	kmol/hr
S11 total	1040.66	kg/hr	57.76631	kmol/hr
S12	Mass Flow Rate	Units	Molar Flow Rate	Units
Water	936.5941	kg/hr	51.98968	kmol/hr
S12 total	936.5941	kg/hr	51.98968	kmol/hr
S13	Mass Flow Rate	Units	Molar Flow Rate	Units
Water	104.066	kg/hr	5.776631	kmol/hr
S12 total	104.066	kg/hr	5.776631	kmol/hr

PFD

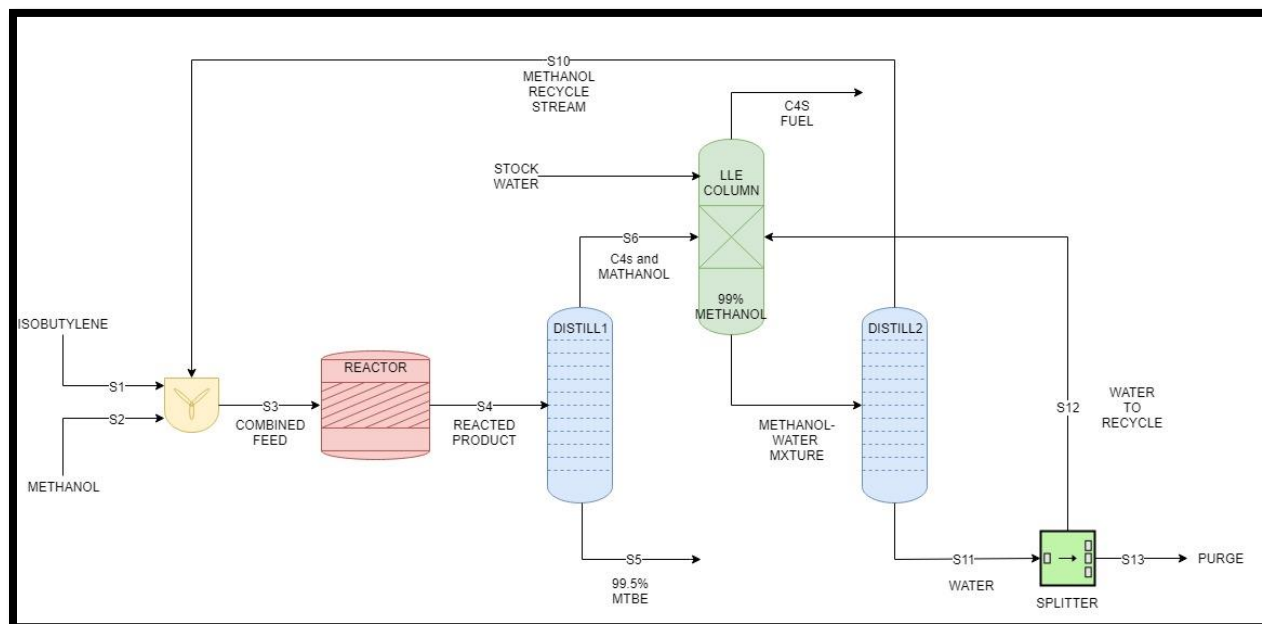


Figure 15 PFD of MTBE Production

Simulation of MTBE Column with Aspen

An Aspen Plus simulation of MTBE distillation column was created and run using first short cut and then rigorous methods to target an achievement of 99.5% MTBE recovery. Streams parameters such as molar compositions, flow rates and reasonably estimated pressure and temperature were first used in short cut method to get inputs for RADFRAC model. Appropriate phase equilibrium and operating conditions were chosen for the rigorous model to reach optimum MTBE recovery by manipulating parameters such as reflux ratio, number of stages, feed location, property method, feed and condenser pressures mostly one at a time to achieve targeted recovery. Furthermore, chemical compositions of the column were graphed over tray positions for a range of different parameters to investigate changes in species behavior in the separation process and MTBE recovery.

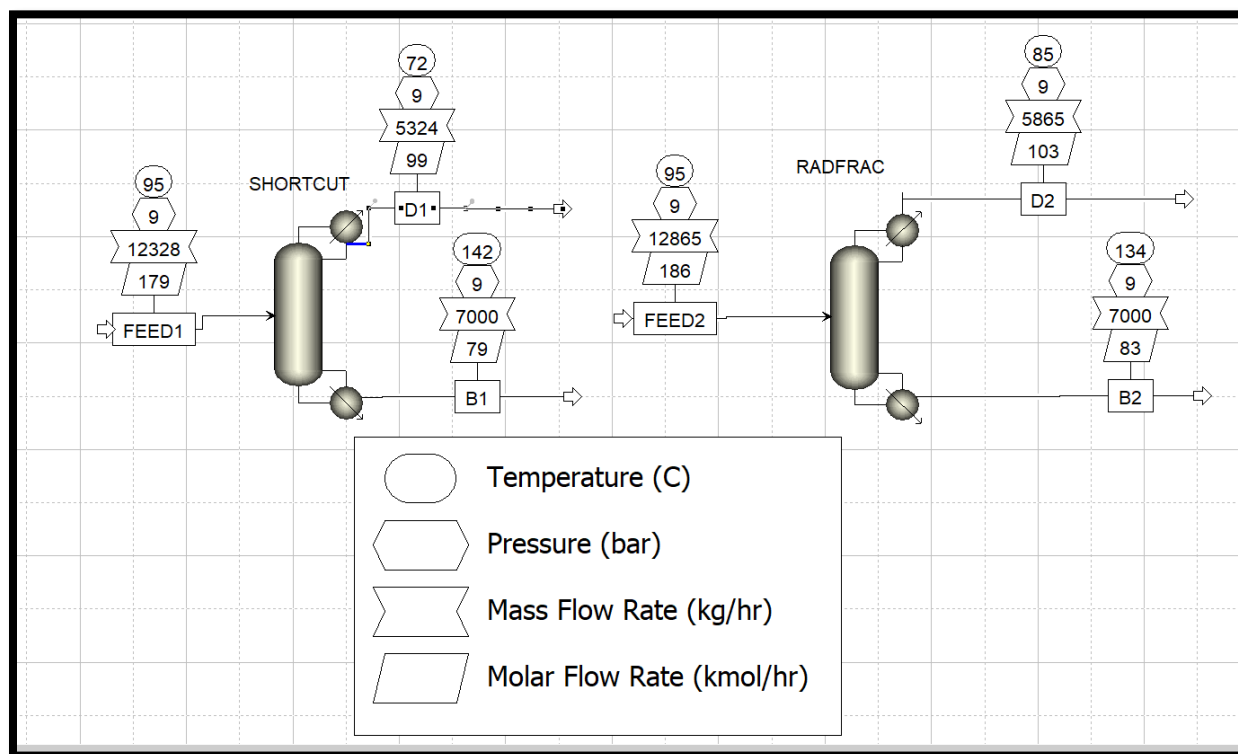


Figure 16 Parallel Diagrams of Shortcut and Radfrac distillation simulation at 9 bars

Table 15 Summary result of short cut simulation using parameters from mass balance

Summary	Balance	Reflux Ratio Profile	Status
▶ Minimum reflux ratio	1.0673		
▶ Actual reflux ratio	1.22739		
▶ Minimum number of stages	12.675		
▶ Number of actual stages	29.94		
▶ Feed stage	20.3122		
▶ Number of actual stages above feed	19.3122		
▶ Reboiler heating required	326004	cal/sec	
▶ Condenser cooling required	287693	cal/sec	
▶ Distillate temperature	72.4665	C	
▶ Bottom temperature	142.306	C	
▶ Distillate to feed fraction	0.554787		
▶ HETP			

Results below are from running the RADFRAC model simulation in Aspen Plus

Parameters used for figure 17:

BASE

Number of Stages	33
Feed at above stages	21
Condenser Pressure	8.6 bar
Reflux Ratio (base)	0.8765

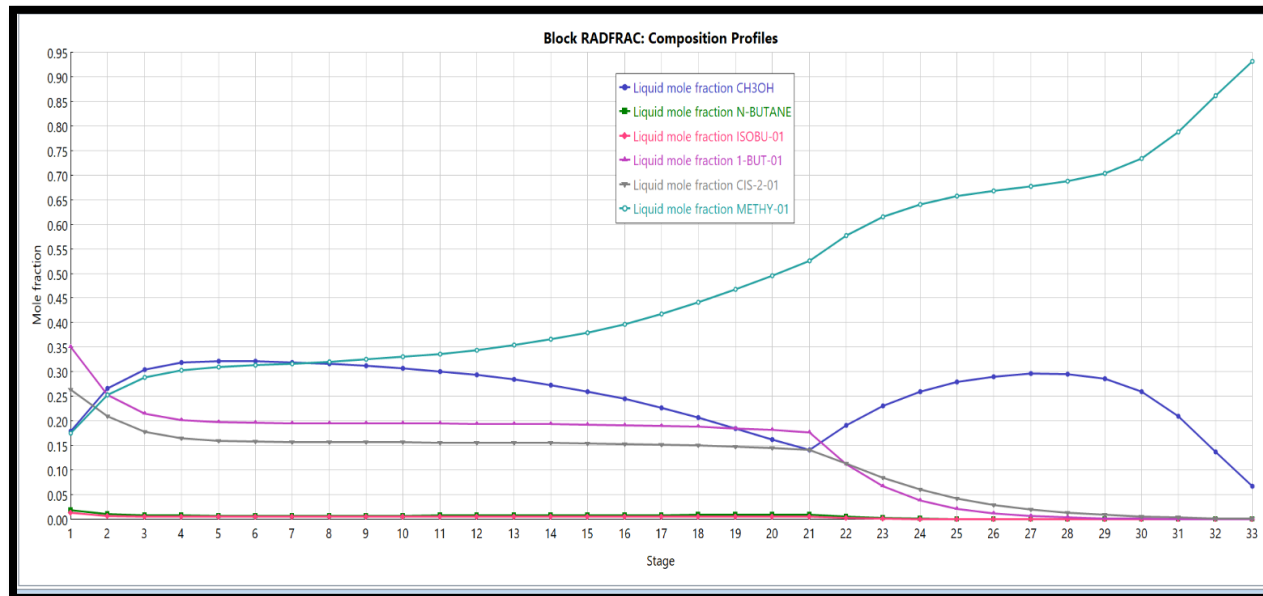


Figure 17 composition vs tray position at base condition

D2: Distillate at the Radfrac model column

B2: Bottom at the Radfrac model column

CH3OH: Methanol

METHY-01: MTBE (Methyl tert-butyl ether)

Table 16 Radfrac Split Fraction results for a base condition

RADFRAC (RadFrac) - Results					Main Flowsheet	
Summary		Balance		Split Fraction		Reboiler
	Component	D2		B2		
▶	CH3OH	0.501075		0.498925		
▶	N-BUTANE	1		3.0529e-06		
▶	ISOBUT-01	1		1.01656e-07		
▶	1-BUT-01	0.99988		0.000119614		
▶	CIS-2-01	0.998004		0.00199645		
▶	METHY-01	0.071403		0.928597		

Figure 17 is a result of implementation of the parameters listed above as a base for a reasonable recovery of MTBE. The figure shows that the mole fractions of species in the distillation column approaches zero as the number of stages increase with the exception of MTBE that rises to around 0.93 as expected given the operating conditions of the column.

Parameters used for figure 18:

Number of Stages	33
Feed at above stages	21
Condenser Pressure	8.6 bar
Reflux Ratio (lower)	0.5

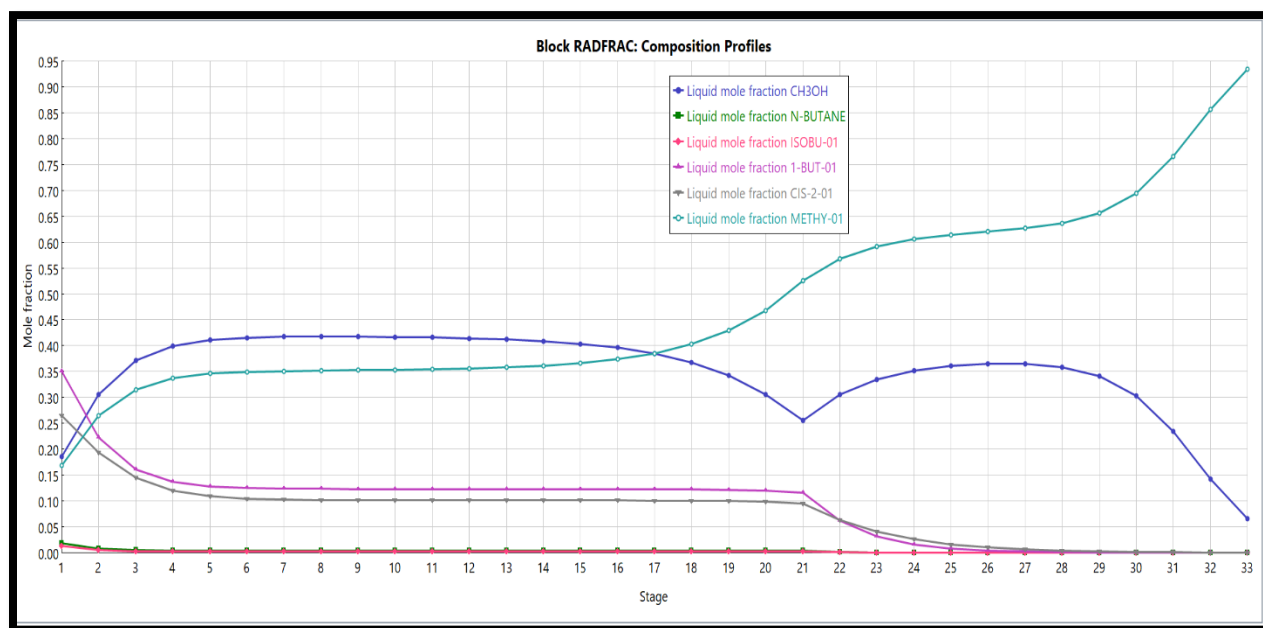


Figure 18 composition vs tray position at lower reflux ratio of 0.5

Table 17 Radfrac Split Fraction results for reflux ratio of 0.5

RADFRAC (RadFrac) - Composition - Plot			
RADFRA			
Summary Balance Split Fraction Reboiler Uti			
	Component	D2	B2
►	CH3OH	0.377423	0.622577
►	N-BUTANE	0.99997	3.00087e-05
►	ISOBUT-01	1	1.2991e-06
►	1-BUT-01	0.99952	0.000480319
►	CIS-2-01	0.992477	0.00752331
►	METHY-01	0.0903096	0.90969

Figure 18 displays the changes in composition and behavior of the species in the column when the reflux ratio is lowered to 0.5 from 0.8765 while keeping all other parameters constant. The effect of a lower reflux ratio on the process is that the recovery rate of MTBE goes down from 92.86% to 90.97% and about 1% change in methanol composition and no significant change in the rest of the species. The decrease in liquid bottom recovery might have to do with the fact that there is a lesser contact of vapor/liquid and operating line for the rectification section moves toward equilibrium line which can be verified using the McCabe-Thiele method.

Parameters used for figure 19:

Number of Stages	33
Feed at above stages	21
Condenser Pressure	8.6 bar
Reflux Ratio (higher)	2.04

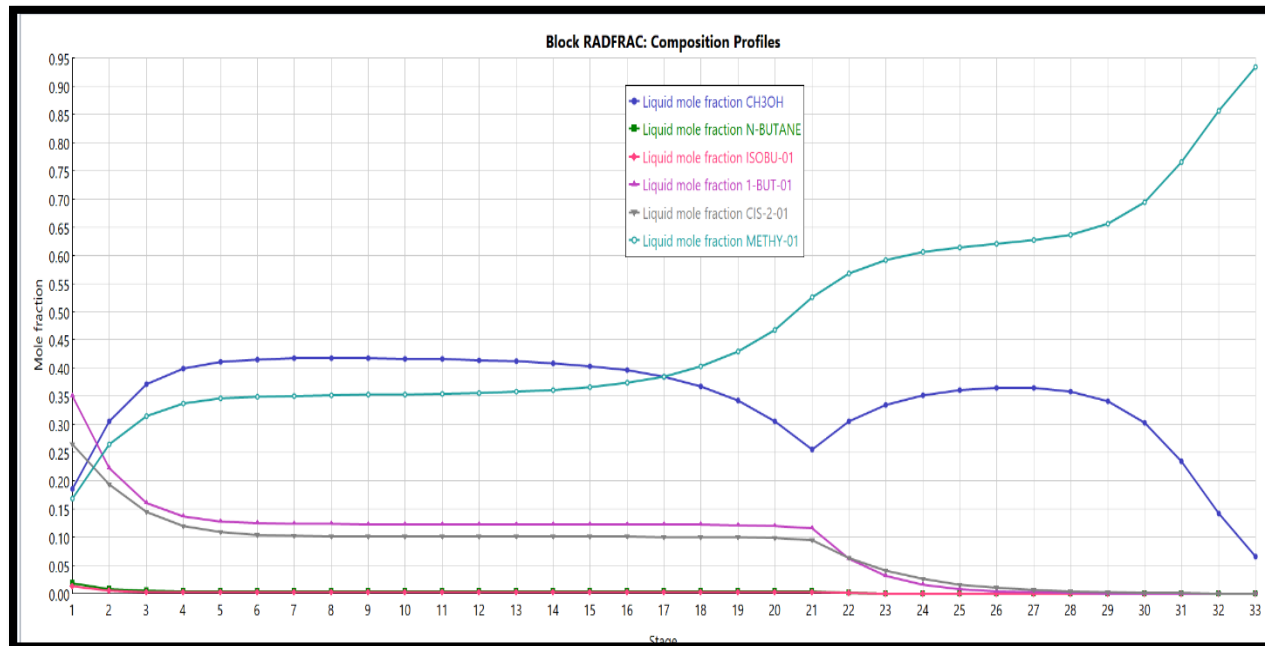


Figure 19 composition vs tray position at higher reflux ratio of 2.04

Table 18 Radfrac Split Fraction results for reflux ratio of 2.04

RADFRAC (RadFrac) - Results				Results Summary	
Summary		Balance	Split Fraction	Reboiler	U
	Component	D2	B2		
▶	CH3OH	0.51339	0.48661		
▶	N-BUTANE	1	9.97182e-08		
▶	ISOBUT-01	1	2.89456e-09		
▶	1-BUT-01	0.999979	2.14742e-05		
▶	CIS-2-01	0.999616	0.00038445		
▶	METHY-01	0.0690774	0.930923		

Figure 19 shows changes in distillation process with a higher reflux ratio of 2 and constant remaining parameters where an increase of MTBE recovery takes place. The MTBE

rate goes from 92.86% to 93.09% while the other species undergo no significant change. The increase of MTBE at higher reflux ratio might have to do with the concept that more and more liquid that is rich in the more volatile components are being recycled back into the column that contributes to an increase of the bottom recovery.

Parameters used for figure 20:

Number of Stages	33
Feed at above stages	21
Condenser Pressure	5 bar
Reflux Ratio (higher)	0.8765

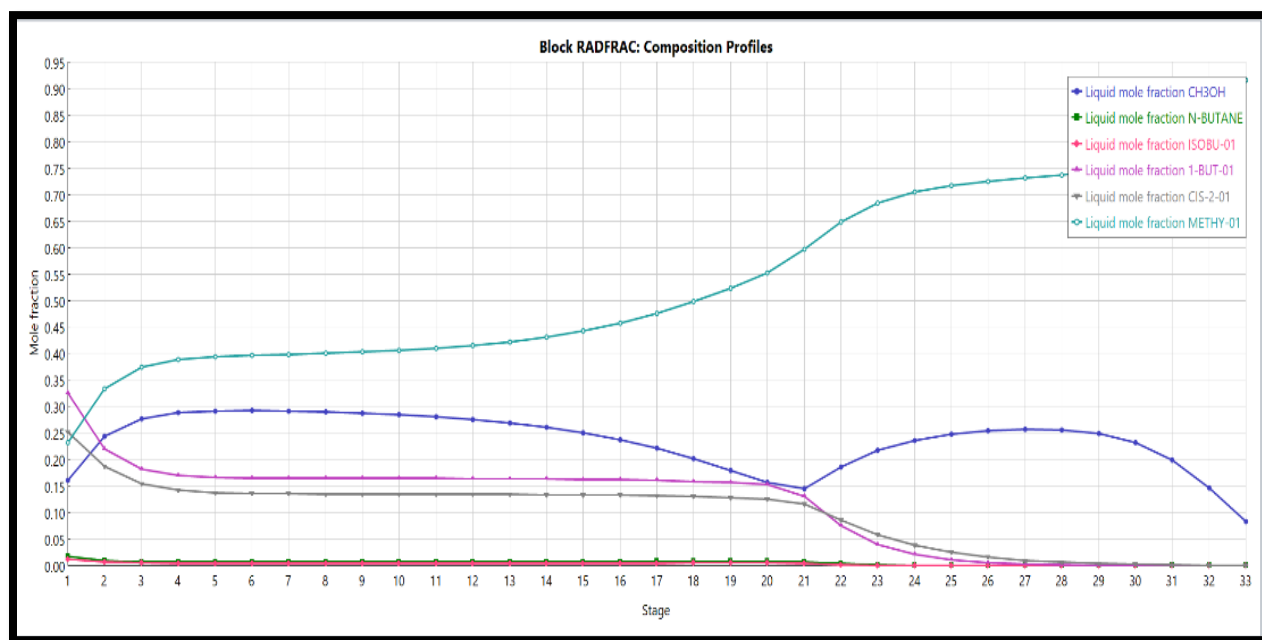


Figure 20 composition vs tray position at lower pressure of 5 bar

Table 19 Radfrac Split Fraction results for pressure of 5 bar

RADFRAC (RadFrac) - Composition - Plot			
RADFRAC			
Summary Balance Split Fraction Reboiler Utili			
	Component	D2	B2
▶	CH3OH	0.378417	0.621583
▶	N-BUTANE	1	1.05701e-06
▶	ISOBUT-01	1	2.9901e-08
▶	1-BUT-01	0.999971	2.90565e-05
▶	CIS-2-01	0.999302	0.000697992
▶	METHY-01	0.0872962	0.912704

Figure 20 displays separation of components at lower pressure where there is a decrease in MTBE recovery and increase in methanol bottom and insignificant change in C4's compositions. Decrease in pressure results in higher boiling point and change in volatility that contributed to the some of the MTBE to go to top part of the column.

Parameters used for figure 21:

Number of Stages	33
Feed at above stages	21
Condenser Pressure	9 bar
Reflux Ratio (higher)	0.8765

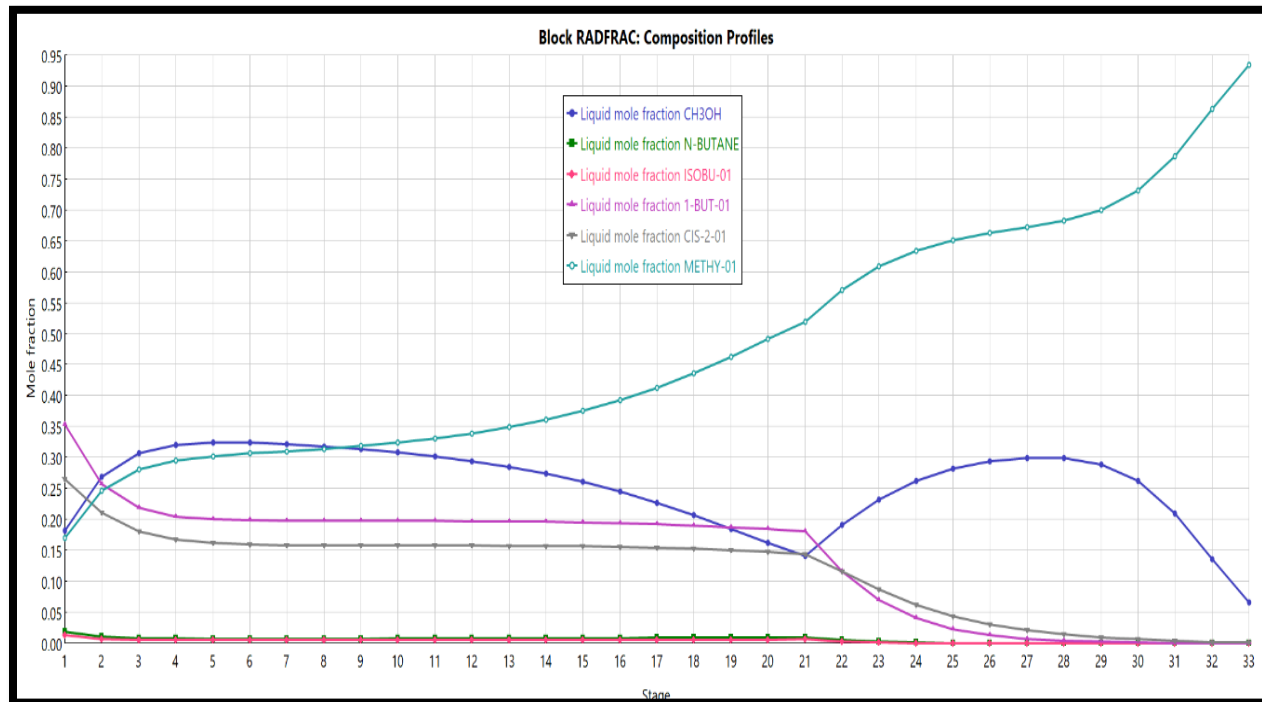


Figure 21 Composition vs tray position at a higher pressure of 9 bar

Table 20 Radfrac Split Fraction results for pressure of 9 bar

Summary Balance Split Fraction Reboiler U			
	Component	D2	B2
►	CH3OH	0.512792	0.487208
►	N-BUTANE	1	3.32389e-06
►	ISOBUT-01	1	1.12227e-07
►	1-BUT-01	0.999867	0.000133446
►	CIS-2-01	0.997837	0.00216329
►	METHY-01	0.0699043	0.930096

Figure 21 shows an increase in MTBE recovery and decrease in methanol composition at the bottom of the column. Increase in pressure leads to decrease in boiling point that favors the

flow of MTBE to the bottom for it is less volatile at this condition. Methanol tends to go to the top in greater composition because it has a higher boiling point at the given pressure.

Parameters used for figure 22:

Number of Stages	43
Feed at above stages	21
Condenser Pressure	8.6 bar
Reflux Ratio (higher)	0.8765

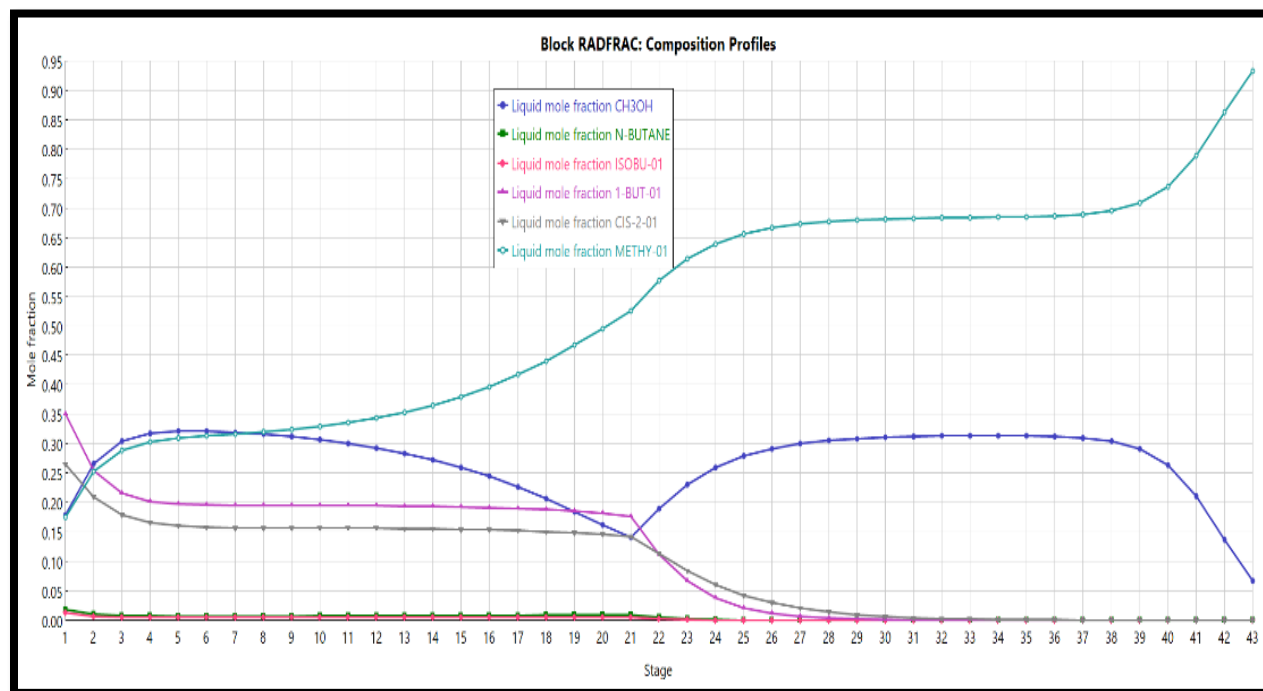


Figure 22 Composition vs tray position at a higher number of stages of 43

Table 21 Radfrac Split Fraction results for number of stages of 43

RADFRAC (Radfrac) - Composition - Plot - Control			
Summary		Balance	Split Fraction
	Component	D2	B2
▶	CH3OH	0.497864	0.502136
▶	N-BUTANE	1	9.28157e-11
▶	ISOBUT-01	1	1.62459e-13
▶	1-BUT-01	1	3.14523e-07
▶	CIS-2-01	0.999955	4.54589e-05
▶	METHY-01	0.0710166	0.928983

Figure 22 displays the changes in species behavior in the separation process with constant parameters except that there is an increase in number of stages. Although increasing the number of stages improves separation by causing the distillate to be much richer in the more volatile component, there is not any significant change in composition of all principal species of the separation. In principle, there should be more methanol at the top and less MTBE at the bottom. The reason that it was not the case might have to do with improper feed position when number of stages increased.

Parameters used for figure 23:

Number of Stages	33
Feed at above stages	11
Condenser Pressure	8.6 bar
Reflux Ratio (higher)	0.8765

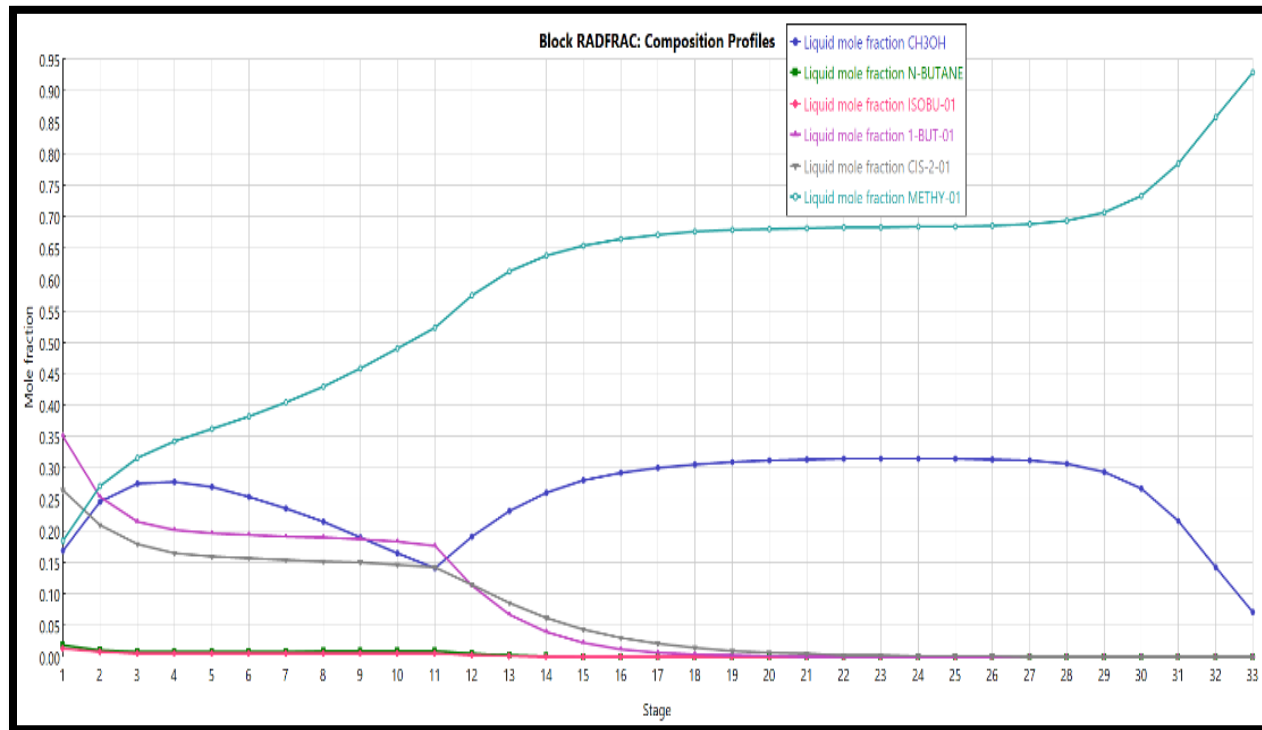


Figure 23 Composition vs tray position at a lower feed stage position

Table 22 Radfrac Split Fraction results for a lower feed stage position

Summary		Balance	Split Fraction	Reboiler
	Component	D2	B2	
►	CH3OH	0.474256	0.525744	
►	N-BUTANE	1	9.14415e-11	
►	ISOBUT-01	1	1.62014e-13	
►	1-BUT-01	1	3.35213e-07	
►	CIS-2-01	0.999952	4.83925e-05	
►	METHY-01	0.0741827	0.925817	

Figure 23 displays changes in species composition in the separation process when the feed is positioned at 10 stages lower. As a result, there is an insignificant or about 0.3 % increase in MTBE and about 0.01 decrease of methanol at the bottom. In principle, lowering a feed stage

makes the top composition less rich in the more volatile component while the bottom richer in more volatile component. Hence, the insignificant changes in the composition of the main separation species might have to with other parameters conditions and operation method such as NRTL in the simulation.

Parameters used for figure 24:

Number of Stages	33
Feed at above stages	21
Condenser Pressure and Feed at 20 bar	20 bar
Reflux Ratio (higher)	2.04

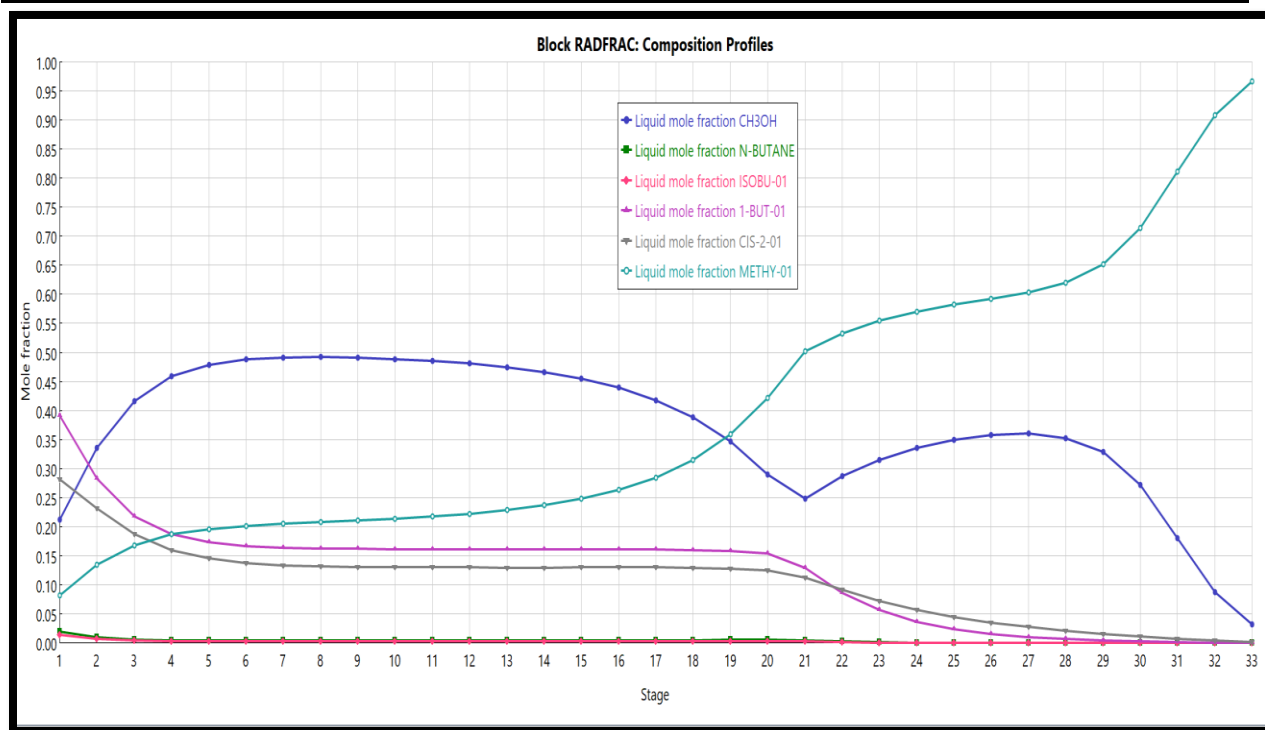


Figure 24 Composition vs tray position at feed and condenser higher pressures of 20 bar and reflux ratio of 2.04

Table 23 Radfrac Split Fraction results for pressures of 20 bar and reflux ratio of 2.04

RADFRAC (RadFrac) - Results			
RADFRAC (RadFrac)			
Summary Balance Split Fraction Reboiler			
	Component	D2	B2
▶	CH3OH	0.762601	0.237399
▶	N-BUTANE	1	3.16433e-06
▶	ISOBU-01	1	7.98051e-08
▶	1-BUT-01	0.999699	0.000300832
▶	CIS-2-01	0.996065	0.00393545
▶	METHY-01	0.0371941	0.962806

Figure 24 displays drastic changes in composition of the main separation components methanol and MTBE when feed and condenser pressures are raised from about 9 to 20 bar and reflux ratio raised from 0.8765 to 2.04. As a result, the recovery of MTBE went up to 96.28% from 92.86% and bottom methanol down to 23.74% from 49.89%. The reason is that a significant increase in feed pressure decreases the boiling point and volatility of MTBE and increase in volatility and boiling point of methanol which make more MTBE to go to the bottom and more methanol to go to the top. Moreover, the increase in reflux ratio leads to increase in purity of distillation when number of stages are constant.

Overall, the application of short cut and then rigorous models in Aspen Plus simulation for the distillation unit of the production of MTBE using variation of parameters, methods and operation conditions lead to a reasonably close MTBE recovery (96.28%) to the actual recovery (99.5%). The properties method that gave the best results is NRTL and the feed and condenser pressure is 20 bar while the reflux ratio is 2.04. All other parameters were taken from material balance of the distillation column and short cut model results. In the column size, the depletion of the C4's components mostly at stage 27 but lesser when pressure and reflux ratio are higher even though the feed position remained the same.

References:

1. Geankoplis, Christie J. Transport Processes and Unit Operations, 3rd Edition. Prentice - Hall International, Inc. 1993
2. Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Editions, Elsevier, 2012.
3. Towler, Sinnott: Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, 2nd Edition Excel Templates. *Elsevier*, booksite.elsevier.com/9780080966595/exceltemplates.php. (accessed Oct 7, 2019)
4. U.S National Library of Medicine. National Center for Biotechnology Information: PubChem. <https://pubchem.ncbi.nlm.nih.gov/> (accessed Oct 7, 2019)